Term paper

Name – Raj kumar

Roll number – 19111044

Semester – 6th

Branch – BME

Subject – DBMS

Topic – students database management system;

ABSTRACT-

An organized and systematic office solution is essential for all universities and organizations. There are many departments of administration for the maintenance of college information and student databases in any institution. All these departments provide various records regarding students. Most of these track records need to maintain information about the students. This information could be the general details like student name, address, performance, attendance etc. or specific information related to departments like collection of data.

All the modules in college administration are interdependent. They are maintained manually. So they need to be automated and centralized as, Information from one module will be needed by other modules. For example when a student needs his course completion certificate it needs to check many details about the student like his name, reg number, year of study, exams he attended and many other details. So it needs to contact all the modules that are office, department and examination and result of students.

With that in mind, we overhauled the existing Student Database Management System and made necessary improvement to streamline the processes. Administrators using the system will find that the process of recording and retrieving students information and managing their classes, including marking of attendance, is now a breeze. In general, this project aims to enhance efficiency and at the same time maintain information accurateness. Later in this report, features and improvement that allow achievement to this goal will be demonstrated and highlighted.

Our work is useful for easy user interface. We are planning to uttilize the powerful database management, data retrieval and data manipulation. We will provides more ease for managing the data than manually maintaining in the documents. Our work is useful for saving valuable time and reduces the huge paper work.

PROJECT REVIEW

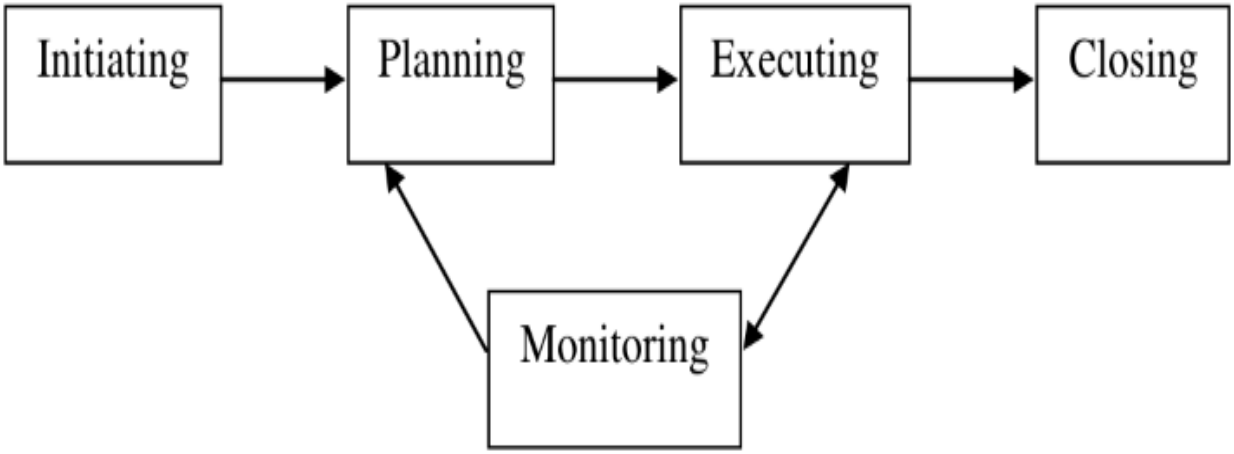

Figure 2.1: Project Development Phases

Project management skills are put to good use for this project. Having gone through project management modules in Time Series Analysis, Optimization and with two interns Project Management for Business and IT respectively, they enhanced my knowledge on managing a project. Project management focuses on achieving the objectives by applying five processes presented in Figure.

RESOURCES-

Systems Development Life Cycle (SDLC) is the most common process adopted to develop a project and not surprisingly, this project is following this model too. To be precise,

waterfall model is being applied. Waterfall model is a sequential model process where the input of a phase actually results from the previous phase.

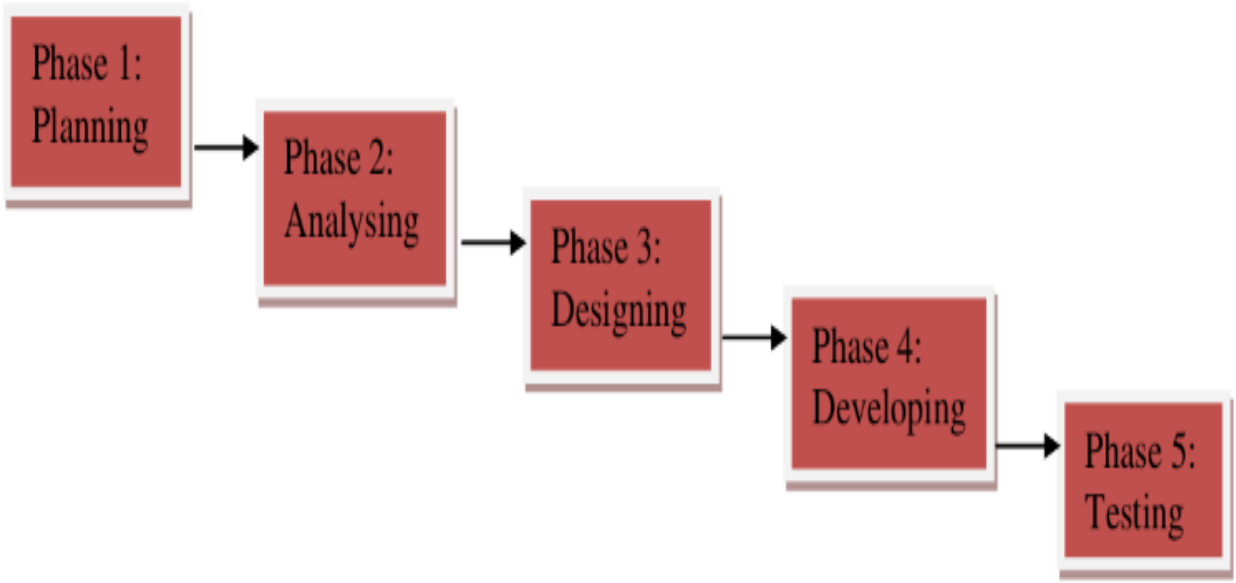

Figure 3.1: SDLC Phases

There are five phases in this model and the first phase is the planning stage. The planning stage determines the objectives of the project and whether the project should be given the green light to proceed. This is where the proposal submission comes into picture. After obtaining the approval, the next phase is analysis. Gathering and analyzing the system and user requirements is essential for entry to the design step.

Package	Release Date	Version	License	Apache HTTP Server	PHP	MySQL	phpMyAdmin	SQLite	Automatic Update
AMPPS	3/8/2011	1.1	Free	2.2.21	Yes	Yes	Yes	Yes	Update Package
WampServer	26/9/2011	2.2a	General Public License	2.2.21	Yes	Yes	Yes	-	-
XAMPP	21/9/2011	1.7.7	General Public License	2.2.21	Yes	Yes	Yes	Yes	-

Figure 3.2: Comparison of web servers

After deciding on the scripting language and database, next is to select the web server that can support them. Web server is necessary for the delivery of web content to the web browser. As such, Apache HTTP server which has performance similar with other 'high-performance' server is considered.

Thereafter, research and actual testing have been performed to see the outcome of the various servers listed in the Figure below. These servers include PHP and MySQL in their installation packages thus allowing smoother and simpler download process. However, based on the performance and interface, Wamp or LAMP server is the preferred choice.

SYSTEM ANALYSIS AND DESIGN-

In order to provide a clearer picture of the functionality provided by the student management module, we have done a use case analysis. Figure below is a use case diagram to present the functionality in the student management module.

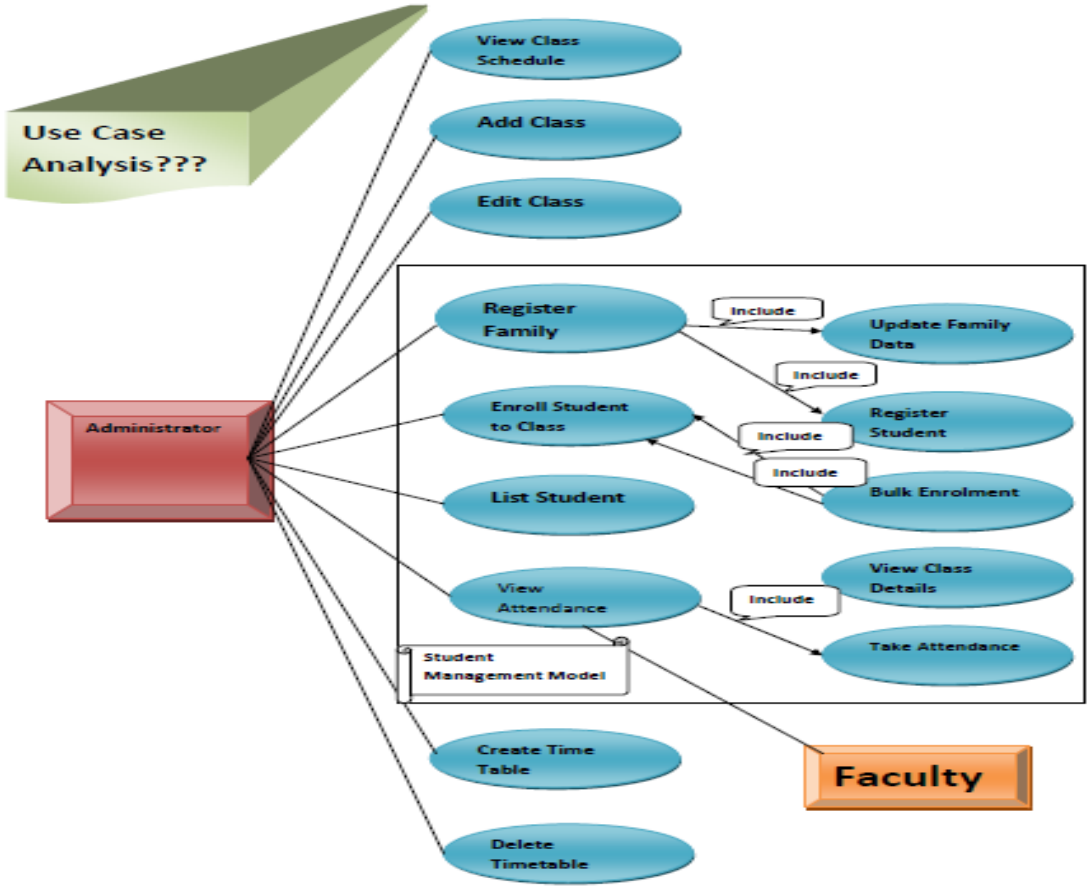

Figure 4.1: use case analysis

Under the existing student management module, the administrator has to first register the family. This includes the parent's particulars and home details. After submitting these data, the administrator will be directed to the student page whereby the student information is to be submitted. Figure below provides the interface of the existing student registration site. For easy navigation, the registration of the student and enrolment to the class has been combined into one process.

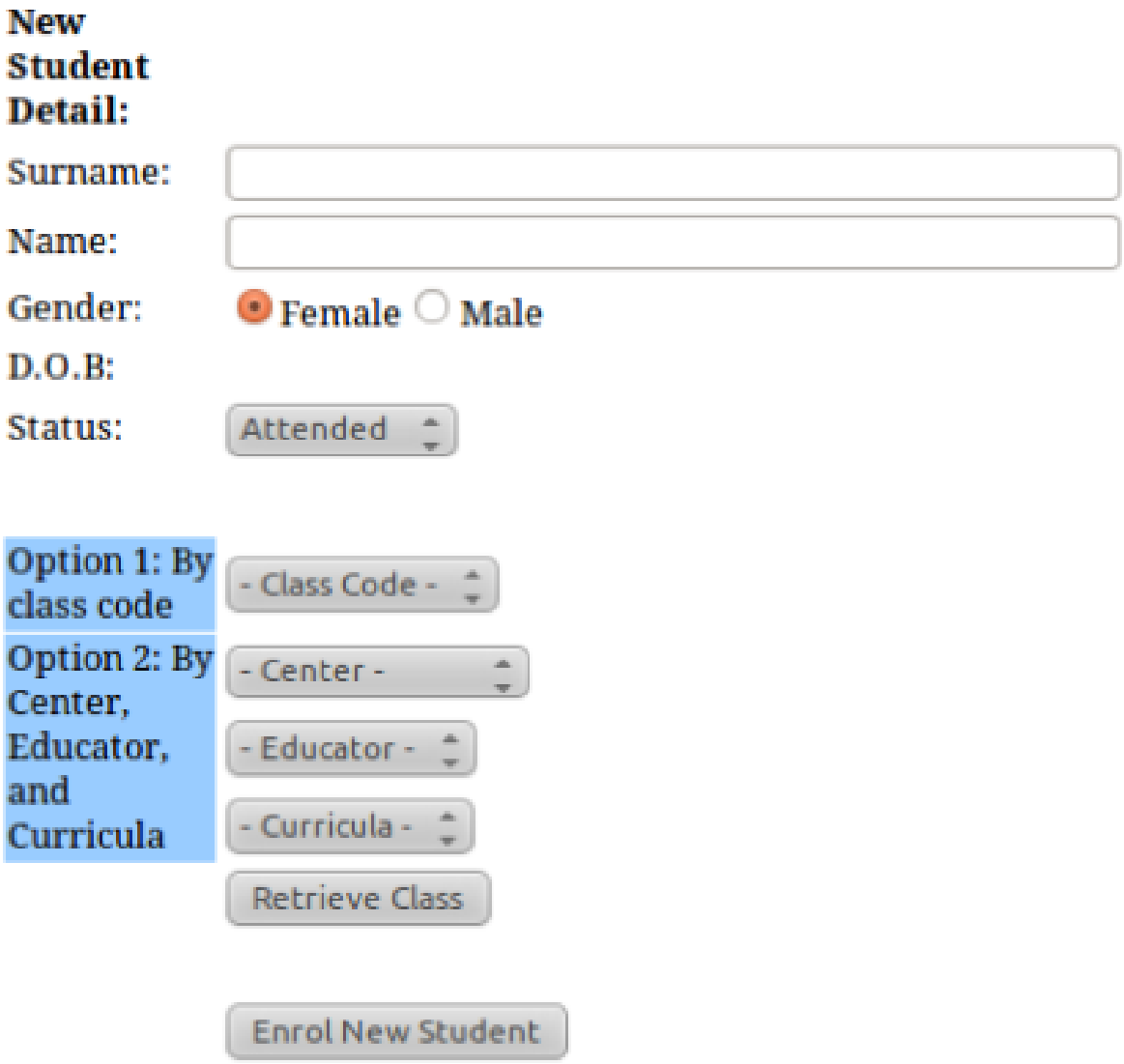

Figure 4.2: Student registration process

DATABASE DESIGN AND DEVELOPMENT

The first step is to create the database as shown in Figure below. The default storage engine in this MySQL server is InnoDB which has commit, rollback, and crash-recovery capabilities to protect user data. Following that is to create the tables in the database.

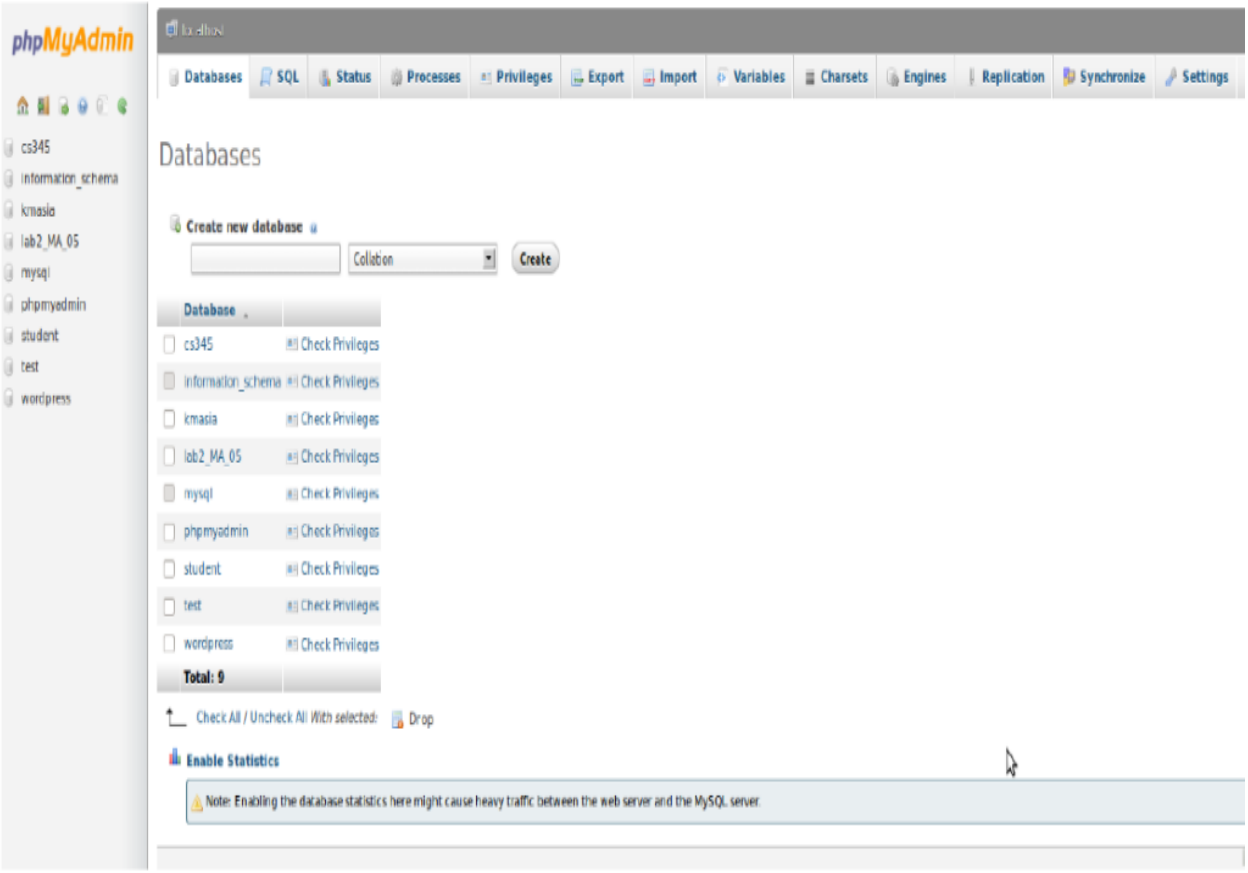

Figure 5.1: Database creation in phpMyAdmin

Figure below gives a view of the table creation.

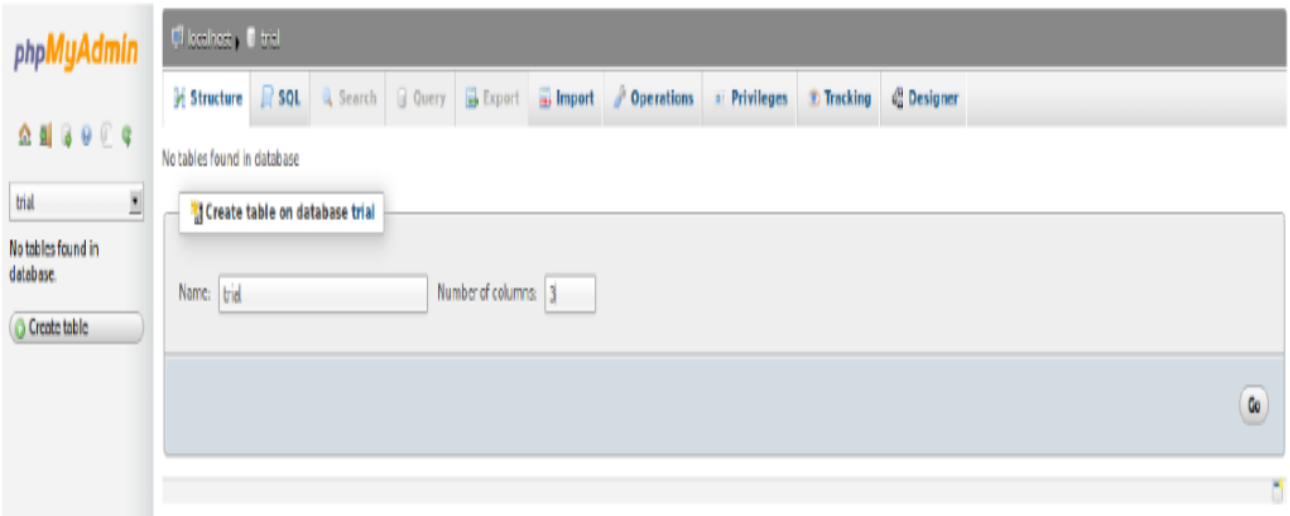

Figure 5.2: Table creation in phpMyAdmin

SYSTEM AND DATABASE TESTING-

Testing is the last phase of this project. This is the stage where it shows whether the system and database are working as desired. Otherwise, further rectification is required. Tests have been conducted and recorded below to show the results.

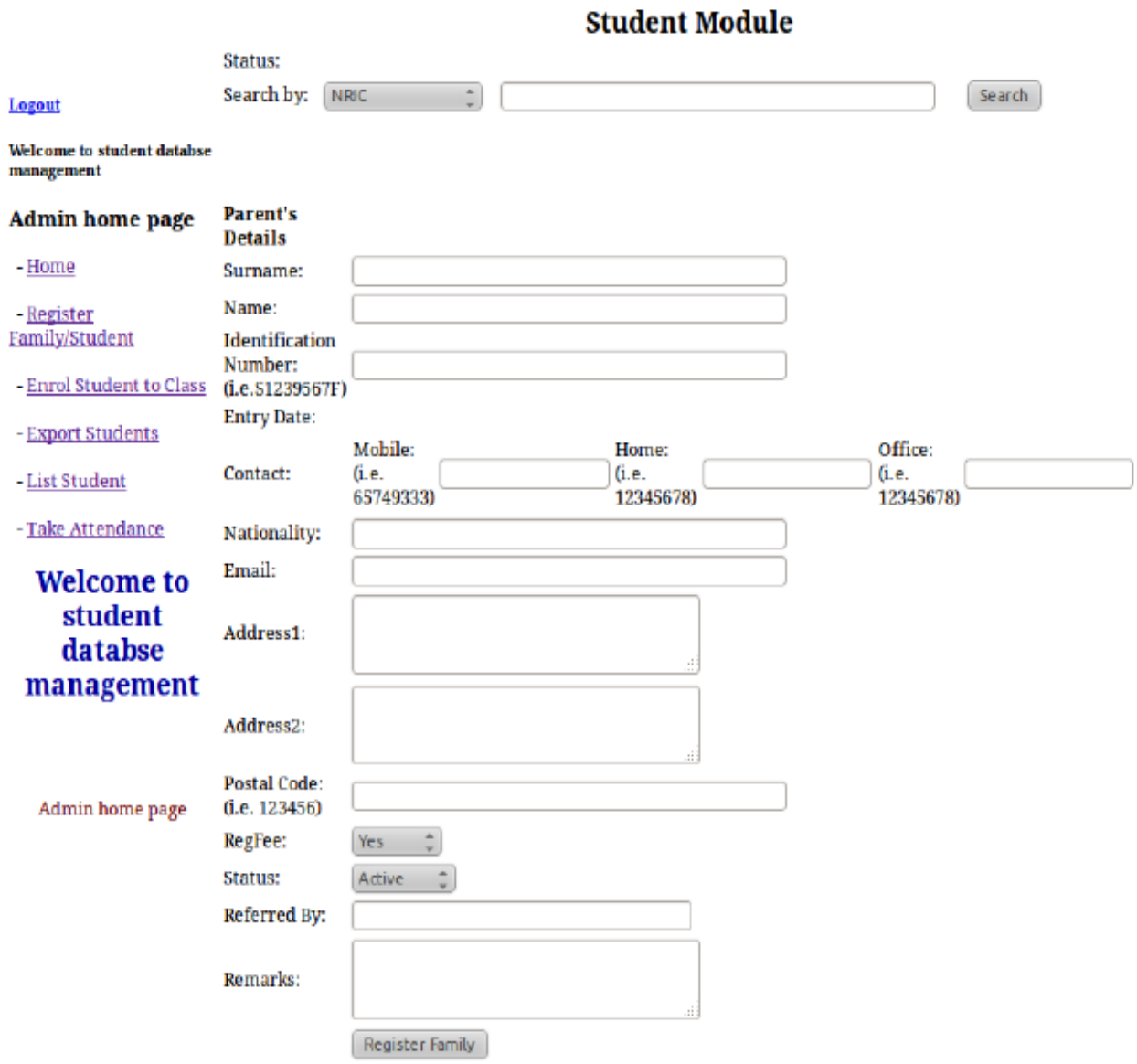

Figure 6.1: Test

PROJECT MANAGEMENT-

Work breakdown structure

The Work Breakdown Structure provides a clear picture of the items that are required to be completed for the project work scope. It is arranged in a tree structure manner

to show all sub-sections required to be accomplished in each phase throughout the project.

Risk management

Every project has uncertainties and they may affect the project success. These uncertainties are known as risk. There is a need to determine the risks involved in the project to reduce the chance of its occurrence as well as develop a plan to either mitigate or avoid the risks that have been identified.

Risk Event	Consequence	Probability	Impact	Risk Response Plan
Incompatible software application with the operating system	Delay in system implementation	Medium	Medium	a) To source for other software applications that is compatible with the operating system b) To find another workstation
User requirements keep changing	More time is required as modification is to be done throughout the whole project period thus unable to meet expected deadline of each phase	Medium	High	a) To check with user on the requirements frequently b) To allocate more time for the development/implementation and testing stage
Difficulties with the coding	Delay in system implementation	Medium	Medium	a) To research more on PHP language and MySQL via sources such as books and tutorial websites
User do not accept the system	Unable to satisfy objectives and complete the project	Low	High	a) Make a check with the user after implementation to ensure the system functioned as per request

Figure 7.1: Risk Assessment Matrix

After identifying and determining the risks, they are recorded in a table called the risk assessment matrix. This matrix lists the severity which is also known as the impact level in terms of High, Medium and Low. In addition to those details, the action plan that can be taken for each of the risks is clearly stated too.

CONCLUSIONS-

Simplicity is never simple. As we have seen in this project, the process of creating a user- friendly and straightforward platform that facilitates the administrator's job is one filled with complexity. From understanding user requirements to system design and finally system prototype and finalization, every step requires in-depth understanding and commitment towards achieving the objectives of the project.

Although the student database management module is not fully integrated to the system and used on real time, the system prototype demonstrates easy navigation and data are stored in a systematic way. Overall, efficiency has improved and work processes simplified. Although all the objectives have been met, the system still has room for improvement. The system is robust and flexible enough for future upgrade using advanced technology and devices.

CRITICAL REVIEWS-

In the process of doing this project, obstacles have been encountered. I have learnt to be more flexible in resolving issues. One of the situations that I have faced was when I had installed MySQL server, but it did not work. Then I tried to troubleshoot, research online and did a couple of re-installation but to no avail. Then I installed XAMPP, it did not work out too. Some how I did get rid of this problem but I did not like XAMPP server. Therefore I decided to install LAMP server. Yes it worked fine. At the end I found that I had installed multiple server and hence system was confused which server is to be used i.e. conflicts with MySQL. Then I stopped all server running on my laptop and decided to start only LAMP server and it worked fine.

--- The end ---